SUPARNO BHATTACHARYYA

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RESEARCH INTERESTS

My research focuses on computational mechanics, reduced-order modeling, structural dynamics, data-driven modeling, mechanical metamaterials, and digital twins. Current work includes projection-based model reduction and hyper-reduction for thermal, structural, and multiphysics systems; data-driven modeling and control of dynamical systems; and machine-learning-based design of mechanical metamaterials. I develop computational frameworks that integrate finite element analysis, model reduction, generative models, and simulation-based design, along with research software in Python, MATLAB, and Mathematica. Prior work includes model reduction of systems with discontinuous and non-smooth forcing, vibro-impact dynamics, topology optimization of lattice structures, hyper-reduced modeling of thermal and neutron transport systems, and experimental characterization of structural damping.

ACADEMIC APPOINTMENT

Assistant Professor, Department of Mechanical Engineering *2025-Present*
Indian Institute of Technology (ISM), Dhanbad, India

PROFESSIONAL EXPERIENCE

Assistant Research Scientist, Digital Twin Lab, Institute of Data Science (TAMIDS) *2023-2025*
Texas A&M University, Texas, USA
Research topic: Reduced order modeling, Digital Twin, Scientific ML.

Postdoctoral scholar, The Department of Automotive Engineering *2022-2023*
Clemson University, South Carolina, USA
Research topic: Data-driven model discovery of dynamical systems, Dynamic modeling of lattice structures for efficient topology optimization.

EDUCATION

PhD, Engineering Science and Mechanics *2021*
Minor: Computational Science
The Pennsylvania State University, PA, USA
Advisor: Prof. Joseph Cusumano
Thesis: Physics-informed Model Reduction of Dynamical Systems Subjected to Impacts and Discontinuity

M.Tech, Mechanical Engineering (Solid Mechanics) *2014*
Indian Institute of Technology Kanpur, India
Advisor: Prof. Anindya Chatterjee
Thesis: Experimental study of damping enhancement in aluminium rods by knurling

BE, Mechanical Engineering *2012*
Jadavpur University, Kolkata, India

FUNDED PROJECTS

- **Texas A&M Institute of Data Science Seed Program for AI, Computing, and Data Science grant**. Start Date: Dec, 2024 (Role: **Lead PI**, $20,000).
  ***Co PIs**: Samuel Gonumakulapalle Lodi (Professor, Indian Institute of Technology Madras), Satish Bukkapatnam (Professor, Texas A&M University, NSF Program Director: ENG/CMMI/AM).*

- **Anusandhan National Research Foundation (ANRF), Prime Minister Early Career Research Grant Scheme**
  Project: Physics-Informed Data-driven Modeling and Uncertainty Quantification with application to Structural Twins. Award Date: Apr, 2026 (Role: **Lead PI**, INR 71,16,720).
  ***Status:** Ongoing*

PRESENTATIONS AND PUBLICATIONS

Journal Publications

J9. **Bhattacharyya, S.**, Syed, A., Tao, J., & Ragusa, J. C. (2026). scikit-rom: An open-source Python platform for teaching and prototyping projection-based reduced-order modeling. arXiv.2608.04960

..... ↑ Journal papers above this line are articles that have gone through at least one round of review and have been accepted contingent upon a revision that needs to be approved ↑.....

J8. **S. Bhattacharyya**, "A 99-Line FEM Code for Simulating a Moving Point Heat Source Over a Domain," Computer Applications in Engineering Education 34 (2026): e70226.

J7. **Bhattacharyya, S.**[†], Tao, J., Gildin, E., Ragusa, J., "Hyper-reduction Techniques for Efficient Simulation of Large-Scale Engineering Systems." Archives of Computational Methods in Engineering (IF: 12.9), 32, p. 5337–5379, 2025.

J6. **Bhattacharyya, S.**, and Cusumano, J.P., "Model reduction of a flexible nonsmooth oscillator recovers its entire bifurcation structure," International Journal of Non-Linear Mechanics, vol. 178, p. 105194, 2025 (IF: 3.2, *Spl. issue: Innovations and Challenges in Non-Smooth Dynamics*).

J5. **Bhattacharyya, S.**[†], Chatterjee, A., "Experimental study of damping enhancement in aluminium rods by knurling." ASME Journal of Applied Mechanics, 92(10): 101003, 2025 (IF: 2.6).

J4. **Khawale, R.**, **Bhattacharyya, S.**[*], Rai, R., Dargush, G., "Efficient dynamic topology optimization of 2D metamaterials based on complementary-energy formulation." Computers & Structures, vol. 299, pp. 107371, 2024 (IF: 4.8, acceptance rate 8%).

J3. **Bhattacharyya, S.**, and Cusumano, J. P., "Experimental Implementation of Energy Closure Analysis for Reduced Order Modeling." ASME, Journal of Vibration and Acoustics, 144(5): 051007, 2022 (IF: 1.9).

J2. **Bhattacharyya S.**, and Cusumano J., "An Energy Closure Criterion for Model Reduction of a Kicked Euler-Bernoulli Beam", ASME, *Journal of Vibration and Acoustics*, 143(4): 041001, 2021 (IF: 1.9, 2.1 at the time of acceptance). ***The paper was selected for the ASME Journal of Vibration and Acoustics spotlight presentation at the 2021 ASME IDETC***.

[†]corresponding author
[*]co-first authors.

J1. **Bhattacharyya S.** and Naskar T.K. (2011), "Analysis of the Effect of Number of Knots in a Trajectory on Motion Characteristics of a 3R Planar Manipulator" *International Journal of Mechanical and Industrial Engineering*, 1(2), pp. 51-56, 2011.

Manuscript Under Preparation

M3. Alsalih, H., **Bhattacharyya, S.**†, Rai, R. "Data-Driven Model Discovery of Flexible Structures."

M2. **Bhattacharyya, S.**, Dargush, G.F., "A Complementary Energy-Based Framework for Efficient Modal Optimization of three-dimensional Parametric Lattice Structures."

M1. **Bhattacharyya, S.**†, Mukherjee, T., Tao, J., "Reduced order modeling of translating heat source for 3D additive manufacturing."

Refereed Proceedings

RP4. Huhn, Q., **Bhattacharyya, S.**, Ragusa, J., "Hyperreduction for Neutron Transport," Transactions of the American Nuclear Society, Volume 131(1): American Nuclear Society Winter Conference and Expo, November 17–21, 2024, Renaissance Orlando at SeaWorld, Orlando, Florida.

RP3. Khawale, R.P., **Bhattacharyya, S.**, Bielecki, D., Rai, R., Dargush, G., "Efficient Methods for Flexibility-Based Meso-scale Dynamic Modeling" in: Allen, M., Blough, J., Mains, M. (eds) Special Topics in Structural Dynamics & Experimental Techniques, Volume 5. SEM 2023. Conference Proceedings of the Society for Experimental Mechanics Series. Springer, Cham.

RP2. **Bhattacharyya, S.**, and Cusumano, J., "The Importance of Energy Criteria for Selecting Modes in Reduced Order Modeling," Proceedings of the ASME 2019 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. Volume 8: 31st Conference on Mechanical Vibration and Noise. Anaheim, California, USA. August 18–21, 2019.

RP1. **Bhattacharyya, S.**, and Naskar, T.K., "The effect of number of knots in a polynomial trajectory on motion characteristics of robotic manipulators," Proceedings of the 9th International Conference on Mechanical Engineering. Organized by Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh. December 18–20, 2011.

Book Chapters

B3. Kumar A., and **Bhattacharyya S.**, "Methods and Materials for Advanced Manufacturing (MMAM) of MEMS/NEMS-Enabled Bio-Electronics and Wearables for Health Monitoring", Basu A.K., Basu A., Ghosh S., & Bhattacharya S. (Eds.), MEMS Applications in Biology and Healthcare, AIP Publishing, Melville, New York, 2021.

B2. **Bhattacharyya S.**, and Kumar, A., "Mechanics of Materials Considerations in MEMS-Based Medical Devices," , Basu A.K., Basu A., Ghosh S., & Bhattacharya S. (Eds.), MEMS Applications in Electronics and Engineering, AIP Publishing, Melville, New York, 2023.

B1. Basu A. K., and **Bhattacharyya S.**, "Scaling Law", Basu A.K., Basu A., Ghosh S., & Bhattacharya S. (Eds.), MEMS Applications in Electronics and Engineering, AIP Publishing, Melville, New York, 2023.

Presentations

P8. **Bhattacharyya, S.**, Dargush, G.F., "Complementary Energy-Based Modal Optimization in Lattice Metamaterials," presented at 7th Indian Conference on Applied Mechanics (INCAM), July 9–11, 2026, IIT Kanpur, India.

P7. **Bhattacharyya, S.**, Tao, J., Bukkapatnam, S., Lodi S.G., Dargush, G.F., "Bandgap Optimization in Symmetry-Driven 3D Lattice Metamaterials," 18th U.S. National Congress on Computational Mechanics (USNCCM18), July 20–24, 2025, Chicago, Illinois.

P6. **Bhattacharyya, S.**, Tao, J., Gildin, E., Ragusa, J., "Uncertainty Quantification With Hyper-Reduced Order Model," ASME Verification, Validation, and Uncertainty Quantification Symposium 2024: May 15-17, 2024, College Station, Texas.

P5. **Bhattacharyya, S.**, Tao, J., Gildin, E., Ragusa, J., "Role of Hyper-Reduction in Enhancing the Efficacy of Digital Twins," 6th Annual Meeting of the SIAM Texas-Louisiana Section (TXLA23): Nov 3-5, 2023, Lafayette, Louisiana.

P4. **Bhattacharyya, S.**, Cusumano, J. "Model Reduction of a Piecewise Nonlinear Mechanical Oscillator," 17th U. S. National Congress on Computational Mechanics: July 23-27, 2023, Albuquerque, New Mexico.

P3. **Bhattacharyya, S.**, **Alsalih, H.***, Choi, Y., Lee, K., Rai, R. "Data-Driven Model Discovery of Flexible Structures," 17th U. S. National Congress on Computational Mechanics: July 23-27, 2023, Albuquerque, New Mexico.

P2. **Bhattacharyya S.**, Cusumano J., "Energy criterion for enhanced accuracy of Reduced Order Models," ESM Today 2018, Penn State University, USA.

P1. Kumar A., Chatterjee K., **Bhattacharyya S.**, "A review on corrosion control methods with electroless coatings," National symposium of micro and nano characterization of materials 2011, Jadavpur University, India. *(Obtained 2nd prize)*

RESEARCH EXPERIENCE

Indian Institute of Technology (ISM) Dhanbad, India *2025–Present*
Assistant Professor, Department of Mechanical Engineering

Research focuses on computational mechanics, data-driven modeling, mechanical metamaterials, generative design, and digital twins. Current work combines mechanics-based simulation with machine learning for design and control of mechanical systems, with research projects conducted through student supervision and collaborative development.

- Supervising research on **data-driven design optimization of mechanical metamaterials for vibration isolation**, with emphasis on computational modeling, structural dynamics, and learning-based exploration of metamaterial design spaces.
- Supervising research on **generative AI for mechanical metamaterial design**, including diffusion-based generative models, simulation-driven data generation, and finite element-based evaluation of generated structures for material and structural design.
- Supervising research on **neural control of mechanical systems**, integrating data-driven dynamical modeling, numerical simulation, and learning-based control toward the development of digital-twin-enabled mechanical systems.
- Developing **data-driven modeling and control strategies for composite structures for flutter suppression**, combining structural dynamics, surrogate modeling, and control methods for aeroelastic response prediction and mitigation.

· Developing computational frameworks that connect mechanics-based models with machine learning for structural design, reduced-order modeling, dynamic response prediction, and control.

Institute of Data Science, Texas A&M University, USA *2023–2025*
Assistant Research Scientist

Research emphasizes the development and application of advanced data-driven modeling and to enhance digital twins across energy, manufacturing, and design domains. Responsibilities include algorithm development, software dissemination, collaborative research leadership, student mentorship, and scholarly contributions.

· Lead developer of open-source software: `pyhyperrom`, a data-driven modeling library for nonlinear systems; currently developing `scikit-ROM`, a finite element–based ROM package for thermal and structural systems.

· Developed data-driven models for heat conduction with time-varying heat sources, representative of additive manufacturing processes.

· Implemented and validated hyper-reduction methods enabling significant computational acceleration for neutron transport simulations (with Prof. Jean Ragusa, Dept. of Nuclear Engineering).

· Principal Investigator on an internally funded seed grant, *AIMS-META: AI-enhanced designing of Manufacturability aware and Symmetry-driven Metamaterials for Enhanced Mechanical Performance* developing machine-learning-based design algorithms that prioritize manufacturability and printability.

· Mentored PhD students on algorithmic research, software development, and scholarly publications.

· Contributed to Lawrence Livermore National Laboratory's open-source ROM software `pylibROM` by extending nonlinear hyper-reduction capabilities and developing benchmark tests.

· Performed Verification, Validation, and Uncertainty Quantification (VVUQ) for hyper-reduced models.

Department of Automotive Engineering, Clemson University, USA *2022-2023*
Postdoctoral scholar

Project #1

Efficient dynamic topology optimization of 2D metamaterials based on a complementary energy formulation (in collaboration with Prof. Gary Dargush at Univ of Buffalo):

· Pioneered a dynamic Topology Optimization framework for printable lattice structures, targeting the attainment of specified dynamic properties.

· Implemented parametric filament-based unit cell structures within the framework to control natural frequency bandgaps effectively.

· Utilized a novel complementary energy-based approach for computing flexibility and stiffness, enhancing computational efficiency by reducing reliance on extensive finite element analysis.

· Demonstrated the framework's efficacy by successfully maximizing band-gap properties in 2D lattice structures through geometric optimization in each cell.

Project #2

Data-driven model discovery of dynamical systems (in collaboration with Prof. Kookjin Lee at Univ of Arizona and Dr. Youngsoo Choi at Lawrence Livermore National Lab):

· Implemented the data-driven modeling techniques: Dynamic mode decomposition and LaSDI (Latent Space Dynamics Identification) for discovering models from structural vibration data.

- Investigated how effectively *neural ODEs* may be integrated with LaSDI for model discovery from vibration data.

Department of Engineering Science and Mechanics, Penn State University *2014 - 2021*
Graduate Research Assistant

Pursued Ph.D. on model order reduction of dynamical systems subjected to impact and discontinuity using data-driven dimension reduction technique Proper Orthogonal Decomposition:

- Performed High-Fidelity structural vibration simulations using FEA and modal analysis in MATLAB.
- Employed Proper Orthogonal Decomposition (POD) on spatio-temporal simulation data to generate computationally efficient reduced order models (ROMs).
- Developed a novel physics-informed criterion for selecting dimension of the ROMs, which yields highly accurate models.
- Explored the application of this criterion for formulating ROMs with experimental data, typically corrupted with measurement noise.
- Demonstrated the effectiveness of physics-informed dimension estimation in generating reliable ROMs for both linear and nonlinear dynamical systems.

Department of Mechanical Engineering, IIT Kanpur *2012 - 2014*
Graduate Research Assistant

Performed experimental study of damping enhancement in aluminum rods by knurling:

- Established that surface deformation processes like knurling can increase structural damping in structural components made of aluminum.
- Engineered an experimental setup to facilitate this study.
- Collected vibration data from the setup utilizing LabVIEW, employing strain gauges and National Instruments DAQ devices for data acquisition.
- Designed both analog and digital filter circuits to mitigate noise in the experimental data.
- Applied optimization techniques to accurately estimate damping parameters from the gathered data.
- Corroborated the findings of increased damping through both theoretical analysis and simulations conducted in ANSYS Fluent.

Department of Mechanical Engineering, Jadavpur University *2008 - 2012*
Undergraduate Research Assistant

Project #1

Studied trajectory planning of planar robotic manipulators

- Showed that increasing the number of control points (knots) while designing trajectory of a 3R manipulator, even though increases its accuracy, can lead to significant fluctuations in velocity, acceleration, and jerk of the manipulator.
- Performed simulation of the motion of a robotic manipulator using AutoLISP language on the AutoCAD platform.

Project #2

Analyzed deformation and deflection of beams with channel cross-section under different loading conditions in ANSYS.

TEACHING EXPERIENCE

Department of Mechanical Engineering, IIT(ISM) Dhanbad *2025-present*
Instructor
• *Research Methodology (NMEC 595)* • *Finite Element Lab (NMEC 535)* • *Introduction to MATLAB and SIMULINK* • *Mechanics for Engineers (NMEE 101)* • *Applied Mechanics Lab (NMEC 205)* • *Mechanical Vibration Lab (NMEC 511)*

Institute of data science, Texas A&M University *2025*
Organizer
TAMIDS Scientific Machine Learning (SciML) Summer School

Free five-day summer school introducing graduate students to the fundamentals of Physics-Informed Neural Networks (PINNs) and Scientific Machine Learning (SciML). The first three days combined morning theory sessions on differential-equation modeling with afternoon hands-on tutorials; the final two days featured invited research talks by leading SciML practitioners. Organized in partnership with the TAMIDS Scientific ML Lab, Los Alamos National Laboratory, and the NASA-DEAP Institute.

Institute of data science, Texas A&M University *2025*
Presenter
Machine Learning in Social Science Workshop

- Delivered a lecture on Principal Component Analysis (PCA), focusing on its use for dimensionality reduction in social science datasets.
- Covered core concepts including variance maximization, eigenvalue decomposition, and data projection; illustrated with Python-based examples.
- Presented to an interdisciplinary audience as part of a workshop organized by TAMIDS Student Ambassadors, in collaboration with the College of Education & Human Development and the Department of Psychological & Brain Sciences.

Institute of data science, Texas A&M University *2024*
Instructor
TAMIDS workshop: Dive into Reduced Order Modeling with pylibROM

Collaborative workshop on Reduced Order Modeling (ROM), organized in partnership with Texas A&M High Performance Research Computing and the libROM team at Lawrence Livermore National Laboratory (LLNL).

Department of Petroleum Engineering, Texas A&M University *2023*
Guest lecturer
PETE 689: Physics-based and Data-Driven Reduced-Order Modeling for Engineering Systems

Offered workshop style lectures on pyMOR, a python library for model order reduction applications, as a part of the course.

Department of Engineering Science and Mechanics, Penn State *2014 - 2021*
Teaching assistant
EMCH 210: Statics and strength of materials, EMCH 212: Dynamics, EMCH 316: Experimental Determination of Mechanical Response of Materials

- Conducted a total of 28 lectures for the course EMCH 212 across two separate sections in the Fall semesters of 2015 and 2016.

- Oversaw and evaluated tests, quizzes, and weekly homework assignments for an average class size of 300 students, coordinating with a team of six teaching assistants.
- Maintained regular office hours and organized weekly tutoring sessions, assisting students in understanding course material, completing assignments, and enhancing academic performance.
- As the lead teaching assistant (TA) for multiple semesters, I mentored new TAs and supervised the grading process during examinations.
- Acquired proficiency in Learning Management Systems such as CANVAS and ANGEL .

Department of Mechanical Engineering, IIT Kanpur *2013 - 2014*
Teaching assistant (TA202: Manufacturing Lab)

- Executed the comprehensive duties of a Teaching Assistant, effectively managing a class of 400 students, ensuring smooth educational operations.
- Diligently assessed and graded project reports, maintaining an up-to-date and accurate record of student grades.
- Provided detailed and insightful feedback to students, guiding them towards academic improvement and deeper understanding.
- Oversaw laboratory sessions, ensuring a conducive learning environment and offering necessary supervision to students.

Self Employed, Pennsylvania, USA *2017 - 2019*
Volunteer – Math and Physics Tutor (private)

Volunteered to tutor two students, one from middle school and another from high school, in the local community, offering personalized educational support.

- Conducted one-on-one tutoring sessions in Advanced Geometry, Advanced Algebra, Pre-calculus, and AP Physics, tailoring instruction to each student's learning style.
- Assisted the students with homework, projects, and preparation for school exams and weekly quizzes, enhancing their academic performance.
- Created customized learning materials to address the unique educational needs and objectives of each student.

CERTIFICATIONS

Summer School: Model Reduction and Machine Learning for Solids, Fluids and Controls *2024*
Centre International des Sciences Mécaniques/International Centre for Mechanical Sciences, Udine, Italy

The summer course introduced participants to the latest advances in nonlinear model reduction, focusing on practical applications. This course explored three main approaches: projection-based, invariant manifold, and neural-network methods, covering both theory and implementation. Lecturers included Professors Balakumar Balachandran, Charbel Farhat, Michael Graham, George Haller, Shobhit Jain, and Gianluigi Rozza.

Summer school: Solving large systems efficiently in multiphysics numerical simulations *2021*
Centre de recherches mathématiques (CRM), Univesity of Laval

A program to introduce the participants to fundamental techniques for solving large multiphysics problems: (i) stationary iterative methods, (ii) domain decomposition, and (iii) multigrid methods, and to well-designed problem sets to experiment with, and to explore the mathematics behind these techniques.

Machine Learning *2020*
Offered by Stanford University via Coursera (online learning platform)

- Supervised learning (parametric/non-parametric algorithms, support vector machines, kernels, neural networks).
- Unsupervised learning (clustering, dimensionality reduction, recommender systems, deep learning).

Graduate Student Online Teaching Certificate *2019*
The Pennsylvania State University, World Campus

A training program to structure and teach an online course through various activities that ensure effective discourse with students. Activities include discussion management, assessment techniques, developing learning activities, reflective practice, and planning for future development and community building.

INVITED TALKS

- Paper titled "Model reduction of a flexible nonsmooth oscillator recovers its entire bifurcation structure", was selected for an invited presentation at the prestigious *Recent Advances in Non-Smooth Dynamics (NSD2025) conference*, organized by European Mechanics Society, held from 8-10th December in Exeter, UK.
- Invited talk at Center for Automotive Research and Tribology IIT Delhi: "Projection based data driven modeling to speed up FEA simulation." (Nov 2025)

HONORS & ACHIEVEMENTS

- Awarded $20,000 Texas A&M Seed Grant for leading a project on AI-driven metamaterial design, selected among 39 participants.
- Received USNCCM18 Travel Award to present research at the 18th U.S. National Congress on Computational Mechanics, 2025.
- ASME Journal of Vibration and Acoustics spotlight presentation at the 2021 ASME IDETC.
- Received the Graduate Student Travel Award from the Department of Engineering Science and Mechanics at Penn State University for the MMLDT-CSET 2021 conference.
- Achieved an All India Rank of 96 among 112,320 candidates in the 2012 Graduate Aptitude Test (GATE) in Mechanical Engineering, a pivotal entrance exam for M.Tech programs at the Indian Institutes of Technology and Indian Institute of Science.
- Attained a rank of 378 in the Engineering category of the State Joint Entrance Examination (WBJEE) in 2008, standing out among approximately 70,000 students.
- Authored *Sangīt Mañjarī: Tagore Ed.* a transliterated collection of Dhrupad compositions for wider accessibility, published on Amazon (2025).

RELEVANT COURSEWORK

Applied statistics, Numerical solution of partial differential equations, Vibration of continuous systems, Nonlinear dynamics, Numerical solution of ordinary differential equations, Finite element method, Concurrent scientific programming (high performance computing), Mathematical methods in engineering, A Hands-on introduction to engineering simulations (Online course offered by Cornell University), Continuum mechanics, Theory of elasticity, Theory of plasticity, Rotor dynamics, Introduction to modern control theory (self-taught).

ACADEMIC OUTREACH ACTIVITIES

Reviewer
Sādhanā

Served as a reviewer for the Springer Sādhanā Journal.

Journal of Vibration and Acoustics

Served as a reviewer for ASME journal of vibration and acoustics.

ICLR 2024 Workshop on AI4DifferentialEquations in Science

Served as a reviewer for ICLR 2024 Workshop on AI4DifferentialEquations in Science, organized by the International Conference on Learning Representations (ICLR) committee.

ASME Manufacturing Science and Engineering Conference

Served as a reviewer for MSEC, 2022 organized by American Society of Mechanical Engineers.

College of Engineering Research Symposium, Pennsylvania State University

Served as the reviewer for CERS, the annual research symposium organized by Engineering Graduate Student Council, Penn State.

Organizer
***Minisymposium** at SIAM-TX-LA 2024, Baylor University*

Organized and chaired a minisymposium on "Data Driven Modeling for Biomechanics Simulations: Innovations and Clinical Applications."

***D2MS2** – Data-driven Modeling and Simulation for Mechanical Systems: virtual seminar series, IIT-ISM Dhanbad*

- Dr. Payel Mukhopadhyay, University of Cambridge on "WALRUS: A Scientific Foundation Model for Fluid Like Continuum Dynamics."
- Prof. Ashesh Chattopadhyay, University of California, Santa Cruz, "On some Theory, Theorems, and Scaling in AI for Science."
- Dr. Shruti Motiwale, Pasteur Labs Inc , "High-speed high-fidelity computational modeling approach for cardiac models."

***Virtual workshops**, Digital Twin Lab, Texas A&M University*

- Prof. Romit Maulik of Pennsylvania State University on "Advancements and Applications in Scientific Machine Learning".
- Dr. Satanik Mukherjee, KU Leuven, Belgium on "Multiscale modeling of articular cartilage: from knee joint mechanics to cellular mechano-transduction for osteoarthritis and tissue engineering applications."
- Prof. Ján Drgoňa of Johns Hopkins University on "Digital Twin Lab Virtual Workshop: NEUROMANCER: Differentiable Programming Library for Data-Driven Modeling and Control"

ESM Today, Pennsylvania State University

Served as the organizing member of ESM Today 2020, Engineering Science and Mechanics department's annual research symposium.

SERVICE AND OUTREACH

Vice president
Society for Indian Music and Arts, Pennsylvania State University

A student-led organization dedicated to the rich and expressive traditions of the performing arts of South Asia.

Committee member
Happy Valley Indian Performing Arts Festival, Pennsylvania State University

Organized by Society for Indian Music and arts.

Judge
Engineering's Got Talent, Pennsylvania State University

Sponsored by the Engineering Graduate Student Council and Engineering Undergraduate Council.

Organizer

Helped organize multiple community events for the local Indian community.

PROFESSIONAL AFFILIATIONS

- Member of *USACM: U.S. Association for Computational Mechanics*

REFERENCES

Prof. Joseph Cusumano
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Director, Texas A&M Institute of Data Science (TAMIDS), Royce E. Wisenbaker Professor, BLOC 227G / WEB 332D, Texas A&M University, College Station, TX 77843

Prof. Gary Dargush
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Department of Mechanical and Aerospace Engineering, 223 Bell Hall, University at Buffalo, Buffalo, NY, 14260.

Prof. Anindya Chatterjee
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